

Supplementary Information

Physics transformers: tensor inputs, law-conditioned attention, and fused reasoning–physics layers for multi-law learning

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SI 1. The operator vocabulary

Table 1 lists all 22 operators in the law vocabulary, with the symbolic meaning used to derive each law signature. The order is fixed; law signatures are dense binary vectors over this order, so reordering would invalidate the stored signatures.

Table 1: The 22-operator law vocabulary.

Index	Operator	Meaning
0	first_time_deriv	∂_t
1	second_time_deriv	∂_t^2
2	first_space_deriv	∂_x
3	second_space_deriv	∂_x^2
4	self_advection	$u u_x$
5	linear_restoring	kx
6	energy_conservation	$E = \text{const}$
7	exponential_relaxation	V/τ
8	nonlinear_trig	$\sin / \cos$ of state
9	gravity_coupling	g
10	algebraic_closed_form	closed-form trajectory
11	piecewise_load	piecewise source
12	separable_modes	$\sin(k\pi x) \sin(l\pi y)$
13	diffusion	νu_{xx}
14	moment_curvature	EIw''
15	laplacian_2d	$u_{xx} + u_{yy}$
16	quadratic_in_space	x^2
17	trig_of_param	$\cos \alpha$
18	angular_velocity_squared	ω^2
19	harmonic	$\sqrt{k/m}$
20	linear_in_space	$(L - x)$
21	poisson_source	$f(x, y)$

SI 2. The quantity vocabulary

The 13 physical-quantity labels used for parameter tokens are: length, force, modulus, inertia, thickness, speed, angle, stiffness, mass, amplitude, resistance, capacitance, voltage. Each domain maps its parameters to a canonical ordered list of quantity ids; beam and cantilever use the identical list (length, force, modulus, inertia, thickness), which is what makes their token sequences identical.

SI 3. Multi-law results: per-domain detail

Table 2 reports medians over 6 seeds per condition (real law signature vs. constant-signature control) for trajectory error, answer error, and physics residual, with paired Wilcoxon p-values. All values are read directly from `multi_law_data.json`.

Table 2: Per-law medians over 6 seeds.							
Law	trajectory err		answer err		phys residual		p (curve)
	cond.	ctrl	cond.	ctrl	cond.	ctrl	
beam	0.047	0.289	0.340	0.404	183	116	0.031
cantilever	0.129	0.252	0.283	0.346	582	1970	0.031
projectile	0.102	0.129	0.110	0.112	0.053	0.043	0.69
pendulum	0.050	0.056	0.021	0.023	0.003	0.004	0.69
spring	0.091	0.088	0.063	0.058	0.154	0.230	0.31
rc	0.099	0.115	0.611	0.509	14.8	10.5	0.16

SI 4. Causal law-swap: per-seed detail

Table 3 gives the per-seed steering indices (median over the 12 pooled samples per seed). The pooled Wilcoxon over the 72 samples gives $p = 2.4 \times 10^{-13}$ for the steering index and $p = 1.7 \times 10^{-13}$ for disruption; the per-seed medians give $p = 0.031$ (6 seeds). Disruption for the control is exactly 0.000 for every seed, every sample.

Table 3: Per-seed steering index (median).							
seed	0	1	2	3	4	5	median
conditioned	-0.429	-0.397	-0.455	-0.420	-0.391	-0.407	-0.413
control	-0.563	-0.589	-0.582	-0.534	-0.605	-0.587	-0.584

SI 5. Few-shot adaptation: per-seed detail

Table 4 reports the three seeds of the few-shot experiment (cantilever, 24 samples, 40 epochs). The generalist was pretrained on the other five laws (96 samples each); the specialist is trained from scratch on the same 24 samples.

Table 4: Per-seed few-shot results.

	seed 0	seed 1	seed 2
generalist (conditioned) answer	0.197	0.287	0.182
generalist (control) answer	0.239	0.437	0.153
specialist answer	0.279	0.580	0.691
generalist (conditioned) curve	0.137	0.134	0.172
specialist curve	0.273	0.284	0.282

SI 6. The two physics channels: full 12-run table

Table 5 lists all 12 runs of the physics-vs-data sweep (heat2d; 2 sample budgets $\times$ 2 physics weights $\times$ 3 seeds). The aggregated means are reported in the main text (Table 5).

Table 5: Physics-vs-data sweep, per run.

samples	w_{phys}	seed	answer err	field max err	field mean err	residual
64	0.0	0	0.0215	0.116	0.024	2.60
64	0.0	1	0.0152	0.207	0.043	4.46
64	0.0	2	0.0245	0.150	0.027	4.04
64	0.05	0	0.0403	0.512	0.151	0.387
64	0.05	1	0.0231	0.496	0.138	0.370
64	0.05	2	0.0187	0.523	0.158	0.589
256	0.0	0	0.0110	0.032	0.007	0.815
256	0.0	1	0.0086	0.033	0.007	0.458
256	0.0	2	0.0125	0.032	0.007	0.347
256	0.05	0	0.0179	0.316	0.083	0.106
256	0.05	1	0.0121	0.270	0.076	0.079
256	0.05	2	0.0150	0.246	0.058	0.061

SI 7. DeepXDE external baseline

Per-instance PINNs (DeepXDE 1.15, PyTorch backend) were trained on three Burgers instances (canonical $\nu = 0.05$; mild $\nu = 0.1$; shock $\nu = 0.02$) for roughly 19 minutes of wall time each (1116–1272 s). The single generalist PHYSFORMER answers all three in one forward pass per instance.

Table 6: Generalist vs. per-instance PINN on Burgers.

	canonical		mild		shock	
	PINN	PHYSFORMER	PINN	PHYSFORMER	PINN	PHYSFORMER
answer err	0.000	0.005	0.000	0.000	0.000	0.037
full-field curve err	0.0003	0.329	0.0006	0.505	0.0003	0.379
phys residual	0.0033	0.0002	0.0013	0.0017	0.0051	0.0016
train wall time	1163 s	—	1116 s	—	1272 s	—

SI 8. Regime theory: exact permutation test

The regime-theory correlation (main text Fig. 5) uses Spearman’s ρ between the a priori equation ambiguity and the measured LCA benefit over the five non-degenerate laws. Because $n = 5$, the null distribution is computed exactly by enumerating all $5! = 120$ permutations of the benefit ranks; $\rho_{\text{obs}} = 1.0$ is achieved only by the identity permutation (one of 60 two-sided configurations), giving the exact two-sided $p = 1/60 = 0.0167$. Including RC (the structurally degenerate law, whose normalized trajectory is parameter-invariant) gives $\rho = 0.77$, $p = 0.10$; RC is excluded from the headline statistic with that rationale stated in the main text.

SI 9. The pre-registered ten-law test: full detail

The pre-registration (`physx/models/ext/pre_registration.json`) was written before any ten-law training. Table 7 gives every per-seed trajectory error, the median benefit per law, the vocabulary alone ambiguity, and the token-identity status. The three falsifiable statements and their outcomes:

- $\rho(\text{amb}, \text{benefit}) > 0$ with $p < 0.05$ over the nine non-degenerate laws: **failed** ($\rho = 0.07$, exact permutation $p = 0.88$).
- Leave-one-law-out predicted benefit ranks correlate with measured ranks: **failed** ($\rho_{\text{OOS}} = -0.58$).
- Pendulum and spring (ambiguity rising to 1.0 in the ten-law set) show positive benefit: **one of two held** (pendulum $+0.54$; spring -0.41).

The learned quantity embeddings do not explain the pendulum anomaly: the maximum cosine similarity between pendulum’s tokens and any other law’s quantity token is 0.30 (computed from the trained checkpoint `lca_real_s0.pt`).

Table 7: **The ten-law out-of-sample test, per seed.** Curve errors: held-out peak-normalized trajectory MAE. Benefit: $1 - \text{median}_s \text{err}_{\text{real}} / \text{median}_s \text{err}_{\text{dummy}}$. Ambiguity is the pre-registered vocabulary-only value; twin marks the only pair whose parameter-token sequences are identical.

Law	real s0/s1/s2	dummy s0/s1/s2	benefit	ambiguity	twin
beam	0.143 / 0.068 / 0.097	0.346 / 0.357 / 0.285	+0.72	1.0	yes
cantilever	0.152 / 0.138 / 0.132	0.208 / 0.209 / 0.248	+0.34	1.0	yes
pendulum	0.042 / 0.054 / 0.076	0.146 / 0.105 / 0.116	+0.54	1.0	no
projectile	0.102 / 0.133 / 0.110	0.128 / 0.108 / 0.129	+0.14	0.5	no
drag	0.059 / 0.090 / 0.067	0.076 / 0.065 / 0.077	+0.11	0.5	no
damped	0.146 / 0.116 / 0.141	0.139 / 0.155 / 0.106	−0.01	0.75	no
rc (deg.)	0.154 / 0.111 / 0.126	0.133 / 0.117 / 0.120	−0.05	0.667	no
kepler	0.171 / 0.151 / 0.157	0.139 / 0.207 / 0.132	−0.13	0.667	no
lc	0.090 / 0.121 / 0.100	0.087 / 0.080 / 0.097	−0.15	0.667	no
spring	0.111 / 0.084 / 0.134	0.068 / 0.101 / 0.079	−0.41	1.0	no

SI 10. Reproducibility

All experiments are driven by committed scripts under `physx/` and read/write committed JSON artifacts under `paper.physformer/fig/`:

- `train_multi.py` — multi-law generalist and dummy control (96 samples/law, 250 epochs, 6 seeds);
- `lca_significance.py` — pooled and per-domain paired statistics from the trained checkpoints;
- `law_swap.py` — at-inference equation swapping (inference only, no retraining);
- `train_fewshot.py` — five-law pretrain, 24-sample adaptation, and the from-scratch specialist;
- `run_physvdata.py` — physics-vs-data sweep on heat2d;
- `regime_analysis.py` — ambiguity computation and exact permutation test (six laws);
- `cache_problems.py` / `regime_oos.py` — ten-law problem sets, pre-registered out-of-sample test;
- `baselines.py` — per-law DeepONet external baseline.

The test suite (`physx/test_physx.py`; 49 physics tests) covers the tensor pipeline, the LCA forward/backward pass, the dummy-mode invariance property, law-swap invariance of the control, the significance aggregation, the ten-law pre-registration, and the DeepONet baseline artifacts. All reported numbers can be reproduced by running these scripts on the committed artifacts; retraining from scratch uses the deterministic seed protocol in the scripts and takes under an hour per run on a 20-core CPU.